

PROJECTION OPERATORS

Projection operators specify the fields returned by an operation.

Name	Description
<code>\$</code>	Projects the first element in an array that matches the query condition.
<code>\$elemMatch</code>	Projects the first element in an array that matches the specified <code>\$elemMatch</code> condition.
<code>\$meta</code>	Projects the available per-document metadata.
<code>\$slice</code>	Limits the number of elements projected from an array. Supports skip and limit slices.

PROJECTION

`$` :

The positional `$` operator limits the contents of an `<array>` to return the first element that matches the query condition on the array. Use `$` in the `projection` document of the `find()` method or the `findOne()` method

when you only need one particular array element in selected documents.

- Create new collection called candidates.
- Upload the dataset [link](#)

Example :

```
test> db.candidate.find({}, {name:1, age:1, gpa:1}).count();
12
test> |
```

VARIATION : EXCLUDE FIELDS

Example:

```
test> db.candidate.find({}, {_id:0, courses:0}).count();
12
test> |
```

\$elemMatch (projection)

The `$elemMatch` operator limits the contents of an `<array>` field from the query results to contain only the first element matching the `$elemMatch` condition.

EXAMPLE : To find candidates enrolled in “Computer Science” with specific

Projection

```
test> db.candidate.find({courses:{$elemMatch:{$eq:"Computer Science"}}},{name:1,"courses.$":1});
[
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae3'),
    name: 'Bob Johnson',
    courses: [ 'Computer Science' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae8'),
    name: 'Gabriel Miller',
    courses: [ 'Computer Science' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064aec'),
    name: 'Kevin Lewis',
    courses: [ 'Computer Science' ]
  }
]
```

[\\$elemMatch: link](#)

- Insert few players using above link.

```
test> db.players.insertOne( {
...   name: "player1",
...   games: [ { game: "abc", score: 8 }, { game: "xyz", score: 5 } ],
...   joined: new Date("2020-01-01"),
...   lastLogin: new Date("2020-05-01")
... } )
{
  acknowledged: true,
  insertedId: ObjectId('6682dbaeec4204f7fecdcdf6')
}
```

```
test> db.players.find({}, {games:{$elemMatch:{score:{$gt:5}}},joined:1,lastlogin:1})
;
[
  {
    _id: ObjectId('6682dbaeec4204f7fecdcdf6'),
    joined: ISODate('2020-01-01T00:00:00.000Z'),
    games: [ { game: 'abc', score: 8 } ]
  }
]
```

\$slice (projection) :

The **\$slice** projection operator specifies the number of elements in an array to return in the query result.

Value

`$slice: <number>`

Description

Specifies the number of elements to return in the `<arrayField>`.

For `<number>`:

- Specify a positive number `n` to return the first `n` elements.
- Specify a negative number `n` to return the last `n` elements.

If the `<number>` is greater than the number of array elements, the query returns all array elements.

`$slice: [ <number to skip>, <number to return> ]`

Specifies the number of elements to return in the `<arrayField>` after skipping the specified number of elements starting from the first element. You must specify both elements.

EXAMPLE :

To retrieve all candidates with first too courses.

```
test> db.candidate.find({}, {name:1, courses:{$slice:2}});
[
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae2'),
    name: 'Alice Smith',
    courses: [ 'English', 'Biology' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae3'),
    name: 'Bob Johnson',
    courses: [ 'Computer Science', 'Mathematics' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae4'),
    name: 'Charlie Lee',
    courses: [ 'History', 'English' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae5'),
    name: 'Emily Jones',
    courses: [ 'Mathematics', 'Physics' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae6'),
    name: 'David Williams',
    courses: [ 'English', 'Literature' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae7'),
    name: 'Fatima Brown',
    courses: [ 'Biology', 'Chemistry' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae8'),
    name: 'Gabriel Miller',
    courses: [ 'Computer Science', 'Engineering' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064ae9'),
    name: 'Hannah Garcia',
    courses: [ 'History', 'Political Science' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064aea'),
    name: 'Isaac Clark',
    courses: [ 'English', 'Creative Writing' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064aeb'),
    name: 'Jessica Moore',
    courses: [ 'Biology', 'Ecology' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064aec'),
    name: 'Kevin Lewis',
    courses: [ 'Computer Science', 'Artificial Intelligence' ]
  },
  {
    _id: ObjectId('6682d28ec38b2a00d2064aed'),
    name: 'Lily Robinson',
    courses: [ 'History', 'Art History' ]
  }
]
```